8. Jenkins a. Create a repository on GitHub and upload project files. b. Install and launch Jenkins (http://localhost:8080). c. Install required plugins (Git, GitHub, Pipeline). d. Create a new Pipeline job in Jenkins. e. Configure Pipeline script from SCM and add GitHub repository URL. f. Add a Jenkinsfile to the repository. g. Configure build trigger (Poll SCM or GitHub Webhook). h. Push code changes to GitHub. i. Verify that Jenkins automatically triggers the build.

✅ Step a: Create GitHub Repository & Upload Files

1. Go to GitHub → **New Repository**
2. Name: jenkins-project
3. Upload files OR use Git Bash:

git init

git add .

git commit -m "Initial commit"

git branch -M main

git remote add origin https://github.com/your-username/jenkins-project.git

git push -u origin main

✅ Step b: Install & Launch Jenkins

1. Download Jenkins, Install and run
2. Open in browser:

   http://localhost:8080

4. Enter admin password
5. Click **Install Suggested Plugins**

✅ Step c: Install Required Plugins

Go to:
👉 Manage Jenkins → Plugins → Available Plugins

Install:

- Git Plugin , GitHub Plugin, Pipeline Plugin

Restart Jenkins if asked

✅ Step d: Create Pipeline Job

1. Click **New Item**
2. Enter name: MyPipeline
3. Select **Pipeline**
4. Click OK

✅ Step e: Configure Pipeline from SCM

1. Scroll to **Pipeline section**
2. Select:
   - Definition → **Pipeline script from SCM**
   - SCM → Git
3. Add Repository URL:

https://github.com/your-username/jenkins-project.git

4. Branch: main

✅ **Step f: Add Jenkinsfile**

👉 In your project folder, create file:

Jenkinsfile

👉 Add this code:

```
pipeline {
  agent any

  stages {
    stage('Build') {
      steps {
        echo 'Building project...'
      }
    }
    stage('Test') {
      steps {
        echo 'Testing project...'
      }
    }
    stage('Deploy') {
      steps {
        echo 'Deploying project...'
      }
    }
  }
```

}

👉 Then push:

git add Jenkinsfile

git commit -m "Added Jenkinsfile"

git push origin main

✅ **Step g: Configure Build Trigger**

Option 1: Poll SCM

1. In Jenkins job → Configure
2. Enable **Poll SCM**
3. Add schedule:

* * * * *

👉 (checks every minute)

Option 2: GitHub Webhook (Best)

1. Go to GitHub repo → Settings → Webhooks
2. Add webhook:

http://localhost:8080/github-webhook/

3. Content type: application/json
4. Select: Just push event

✅ **Step h: Push Code Changes**

git add .

git commit -m "Updated code"

git push origin main

✅ **Step i: Verify Auto Build**

👉 Go to Jenkins dashboard

✔ You will see:

- Build triggered automatically
- Console Output showing:
    - Building...
    - Testing...
    - Deploying...